The Final Remainder of Resolution Geometry

Exclusion of a Primitive, Classification of the Coherent Global Structure, and the Boundary at the Walls

Universal properties, natural versus conditional statements, and a hostile falsification battery

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Abstract

After the information residual of Resolution Geometry (RG) was locked relative to a declared contract, one question remained: is the final remainder a *new primitive*, or is it a *global/coherent structure* of a classical kind? We answer it on an explicit domain and mark exactly where the answer stops. For RG families of locally constant *multiplicity type* over a paracompact manifold S (declared resolved/null splitting, covariance tensors, structure-preserving bundle isomorphisms) we prove a *Global Structure Theorem*: the groupoid of such families is equivalent, naturally in S, to the groupoid of tuples $(\sigma; E_1, \dots, E_k, K_R, K_N)$, where $\sigma = (\sigma_1 > \cdots > \sigma_k)$ are smooth canonical-correlation functions and E_j, K_R, K_N are Euclidean vector bundles of ranks m_j, $p-r$, $q-r$. Consequently the remainder splits canonically into *local moduli* $\sigma \in C^\infty(S, \Delta_k)$ and a *global class* $[E_\bullet, K_R, K_N] \in \prod_j [S, B\,\mathrm{O}(m_j)] \times [S, B\,\mathrm{O}(p-r)] \times [S, B\,\mathrm{O}(q-r)]$; this pair is the initial complete natural invariant, hence unique up to unique isomorphism, and its homotopy part is represented by a classifying space unique up to homotopy equivalence. The pointwise automorphism group is $\prod_j \mathrm{O}(m_j) \times \mathrm{O}(p-r) \times \mathrm{O}(q-r)$. This refutes the corpus claims that the coupling rank controls the symmetry dimension and that full-rank coupling rigidifies to the trivial automorphism (witness $C = \frac{1}{2} I_3$: rank 3, $\dim \mathrm{Aut} = 3$), and it refutes the reading of the residual as an intrinsically nonzero descent class (every family over a contractible base is globally trivial) and as an obstruction to a faithful functor (forgetful functors are faithful). A second level governs gluing when transitions are known only up to an abelian automorphism band: the triple defect defines a class in $\check{H}^2$, removable iff zero (removable witness on a filled triangle; non-removable witness on $\mathbb{R}P^2$). At type-changing walls the twisted-form description fails on every neighbourhood, and a transversal collision of two simple canonical correlations *forces* a nonzero first Stiefel–Whitney class around the wall. The verdict is: **no primitive**; **the final remainder is coherent global structure**, locked on constant-type strata and explicitly open across walls. Every claim is attacked by an executable battery (18 attacks, 4 canaries, master seed 20261022).

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1 Introduction

1.1 The question and its two admissible answers

The previous lock established three set-level facts relative to a declared contract, written $\mathcal{K} = (X, c, (A_i)_i, M)$. (i) the information residual is the family $\mathcal{R}_y = \{a(x) : c(x) = y\}$ of jointly attainable task outputs over conversion fibres, and the tasks descend through c iff every $\mathcal{R}_y$ is a singleton; (ii) the repair $E = \{(c(x), a(x))\}$ is the coarsest sufficient refinement; (iii) operational stability is the coarsest observation-preserving M-stable equivalence. It left open whether the *final* remainder of RG—what survives after all local repairs—is

(a) a *new primitive*: an object not obtainable from the declared data by established constructions, or

(b) a *global/coherent structure*: an object obtained by established constructions (descent, cohomology, classifying spaces) and characterized by a universal property.

The corpus contains statements of both kinds. The Lock Conclusion asserts that RG "admits no faithful functor into any single foundation" and that the residual is "a nonzero descent class—RG is a nontrivial bundle of foundations"; Global_RG proposes that "local similarity type + Čech cocycle class = static globalization data". This paper decides between (a) and (b) on an explicit domain, proves uniqueness and universality of the answer, and names the exact boundary beyond which neither is proved.

1.2 What the battery found

Each load-bearing claim was attacked by an executable witness before being used (§15). Five corpus claims are refuted as stated, six survive only in sharpened form, five hold on one side of an explicit hypothesis, two are confirmed on their stated domain. Table 3 (§13) lists them. The decisive findings are:

- the automorphism group is controlled by *multiplicities* of canonical correlations, not by the coupling rank; the corpus rigidity proposition was supported by a random sweep that cannot see a measure-zero stratum (A05);
- a nonzero global class is a property of a *family over a base*, not of RG; over a contractible base it vanishes (A02), and over S^1 it can be nonzero even when every local invariant is constant (A03) and even when $\dim \mathrm{Aut} = 0$ (A06);
- a Čech triple defect is an obstruction only through its cohomology class (A08);
- at walls a global remainder is *forced* by the geometry of the collision (A15).

1.3 Established branches, their overlap, and the gap

The RG remainder lives where several established theories meet. Each supplies a universal construction under a hypothesis that the others do not supply. The gap is not missing mathematics; it is unverified *hand-offs* between branches.

B1. Sufficiency and comparison of experiments. Fisher–Neyman factorization and the Lehmann–Scheffé minimal sufficient statistic [14] identify the coarsest partition that preserves inference; Blackwell and Le Cam [2, 13] order experiments by garbling. These are fibrewise and measure-level. They produce no gluing data and do not see the topology of a parameter family.

B2. Category theory of quotients. Kernel pairs, coimages and coequalizers [17] give the exact extensional layer (Ladder R1–R4). Terminal objects are unique up to unique isomorphism; this is the source of every uniqueness statement below. Category theory alone does not supply regularity or topology.

B3. Coalgebra and automata. Myhill–Nerode [21], bisimulation [18] and partition refinement [22] give the coarsest operation-stable congruence. They presuppose operations as total endomaps and are blind to topology.

B4. Matrix analysis and canonical correlation. Schur complements [30], canonical correlations [9] and the singular value decomposition classify a single covariance under block-linear changes. They presuppose a fixed splitting and say nothing about families.

B5. Descent, stacks and nonabelian cohomology. Grothendieck–Giraud descent [6, 27] classifies objects locally isomorphic to a model by $\check{H}^1$ with automorphism coefficients (twisted forms [24]); gerbes and $\check{H}^2$ [6, 4, 5] govern the existence of global objects from local ones. Descent presupposes a *stack* of local objects with constant local type; it does not produce the local objects.

B6. Fibre bundles and classifying spaces. Principal bundles, reduction of structure group and classifying spaces [25, 10, 19] identify $\check{H}^1$ with $[S, BG]$; holonomy and Ambrose–Singer [1, 12] describe connections. These presuppose a single structure group, i.e. constant isotropy.

B7. Stratified spaces and perturbation theory. Whitney stratifications [29, 23], Luna slices [15], Kato–Riesz spectral projections [11], the von Neumann–Wigner non-crossing rule [28] and the Herzberg–Longuet-Higgins sign change [8] describe what happens when isotropy or spectral multiplicity jumps. Exit-path and constructible-stack formalisms [26, 16] are the natural candidates for a global classification across strata.

The hand-offs. RG statements routinely cross from the guarantee of one branch to the hypothesis of another. Table 1 lists every crossing used in this paper and the result that closes it.

	Hand-off	**Hypothesis required**	**Closed by**
H1	sufficiency (B1,B2) → descent (B5)	local repairs form a stack	Prop. 5.2
H2	set-level → smooth descent	surjective submersion	Prop. 3.7
H3	local symmetry (B4) → principal bundle (B6)	locally constant multiplicity type	Thm. 6.4, Prop. 9.1
H4	Čech on a cover → cohomology of S	Leray (good) cover	Prop. 8.6
H5	holonomy matrices → invariants	conjugacy invariance	Prop. 8.4
H6	coupling block → invariant	declared complement N	Prop. 4.2
H7	random sweep → universal claim	enumeration of strata	Rem. 4.6
H8	triple defect → obstruction	class in $\check{H}^2$, band abelian	Prop. 8.1

Table 1: Hand-offs between established branches and the result closing each.

1.4 Natural versus conditional: the discipline used throughout

A proof that holds after a choice, or under a hypothesis, is not a proof of a natural statement. We tag every result.

Definition 1.1 (Natural, conditional, choice-dependent)**.** Let $\mathcal{D}$ be a category (or groupoid, or stack) of declared data with declared morphisms and base changes.

(i) A construction F on $\mathcal{D}$ is *natural* [N] if it is defined without auxiliary choices and is a (pseudo)functor: every declared morphism and base change f induces $F(f)$, compatibly with composition up to coherent isomorphism.

(ii) A statement is *conditional* [C: H] if its proof uses a hypothesis H not implied by the declared data.

(iii) A construction is *choice-dependent* [X: c] if its value depends on an auxiliary choice c (frame, complement, reference model, cover). Only its class under change of c is meaningful, and it is *promoted* to a natural construction only by an explicit transformation law showing invariance.

Remark 1.2. Promotion can fail. The coupling coordinate $L = CD^{-1}$ is choice-dependent and *not* promotable: under a change of complement it moves affinely (Prop. 4.2). A holonomy matrix is promotable to its conjugacy class (Prop. 8.4). A Čech class on a cover is promotable to $\check{H}^1(S)$ only under Leray refinement (Prop. 8.6).

1.5 Main result

Fix integers $p, q \geq 0$ and a *multiplicity type* $\mu = (p, q; m_1, \ldots, m_k)$ with $r = \sum_j m_j \leq \min(p, q)$. Write $\Delta_k = \{\sigma \in \mathbb{R}^k : 1 > \sigma_1 > \cdots > \sigma_k > 0\}$.

Main Theorem (Final Remainder Lock). *Let S be a paracompact smooth manifold and let $\mathrm{RG}_\mu(S)$ be the groupoid of RG families of multiplicity type μ over S (Definition 5.1). Assume (H-split) the resolved/null splitting is declared, (H-type) the multiplicity type is locally constant, and (H-mor) morphisms are bundle isomorphisms over id_S preserving all covariance tensors.*

(1) ***Exhaustion*** [N]. *There is an equivalence of groupoids, natural in S,*

$$\Phi_S : \ \mathrm{RG}_\mu(S) \xrightarrow{\simeq} C^\infty(S, \Delta_k) \times \prod_{j=1}^{k} \mathrm{Euc}_{m_j}(S) \times \mathrm{Euc}_{p-r}(S) \times \mathrm{Euc}_{q-r}(S),$$

where $C^\infty(S, \Delta_k)$ is regarded as a discrete groupoid and $\mathrm{Euc}_m(S)$ is the groupoid of rank-m Euclidean vector bundles. Two families are isomorphic iff their local moduli σ coincide and their block bundles are pairwise isomorphic.

(2) ***Universality and uniqueness*** [N]. *The invariant $y \mapsto (\sigma(y), [E_\bullet(y)], [K_R(y)], [K_N(y)])$ is the initial complete natural invariant of RG families (Theorem 7.2); it is unique up to unique natural isomorphism. For fixed σ its global part is represented on CW bases by $B\,\mathrm{Aut}_\mu \simeq \prod_j B\,\mathrm{O}(m_j) \times B\,\mathrm{O}(p-r) \times B\,\mathrm{O}(q-r)$, unique up to homotopy equivalence.*

(3) ***Independence*** [N]. *Neither component determines the other: there are families with equal moduli and different global classes, and families over a point with different moduli.*

(4) ***Exclusion of a primitive*** [C: H-split, H-type, H-mor]. *Every component of the remainder, including the information layer of the previous lock, is $\mathfrak{T}$-derived (Definition 10.1); no new primitive is required.*

(5) ***Second level*** [C: abelian band]. *If only local families with transitions defined modulo an abelian automorphism subsheaf $\mathcal{A}$ are given, a global family exists iff a canonical class in $\check{H}^2(S; \mathcal{A})$ vanishes.*

(6) ***Boundary*** [N] *(negative) and* [C: transversality] *(wall). If the multiplicity type jumps at s_0, no neighbourhood of s_0 carries a twisted-form description by a single model. A transversal collision of two simple canonical correlations forces $w_1 \neq 0$ of each colliding singular line bundle on every small loop linking the collision locus. The classification across walls is open.*

Verdict. The final RG remainder is a global/coherent structure of classical type, not a primitive. It is locked on constant-type strata. Across walls it is witnessed but not classified.

2 Conventions

All manifolds are smooth, Hausdorff, second countable (hence paracompact). Vector bundles are real and smooth. For a vector bundle V, $\mathrm{Sym}^2 V$ denotes symmetric 2-tensors; a section of $\mathrm{Sym}^2 V$ is positive if it is positive definite on V^*. A *Euclidean* bundle is a vector bundle with a smooth fibre inner product; morphisms of Euclidean bundles are fibrewise isometric isomorphisms.

Definition 2.1 (Prestacks, stacks, twisted forms)**.** A *prestack in groupoids* $\mathcal{F}$ over S assigns to each open $U \subseteq S$ a groupoid $\mathcal{F}(U)$ and restriction functors compatible up to coherent isomorphism. It is a *stack* if (S1) for $x, y \in \mathcal{F}(U)$ the presheaf $\mathrm{Isom}(x, y)$ is a sheaf, and (S2) every descent datum $(x_a \in \mathcal{F}(U_a),\ \varphi_{ab} : x_b|_{U_{ab}} \to x_a|_{U_{ab}})$ with $\varphi_{ab}\varphi_{bc} = \varphi_{ac}$ on U_{abc} is effective. For $x \in \mathcal{F}(S)$ a *twisted form* of x is $y \in \mathcal{F}(S)$ locally isomorphic to x; $\mathrm{Forms}(x)$ is the set of isomorphism classes of twisted forms, pointed by $[x]$. Aut_x denotes the sheaf of groups $U \mapsto \mathrm{Aut}(x|_U)$.

Definition 2.2 (Nonabelian Čech $\check{H}^1$)**.** For a sheaf of groups $\mathcal{G}$ and an open cover $\mathcal{U} = (U_a)$, a 1-cocycle is $g_{ab} \in \mathcal{G}(U_{ab})$ with $g_{ab}g_{bc} = g_{ac}$ on U_{abc}; $g \sim g'$ iff $g'_{ab} = h_a^{-1} g_{ab} h_b$ for some $h_a \in \mathcal{G}(U_a)$. $\check{H}^1(\mathcal{U}; \mathcal{G})$ is the pointed set of classes, and $\check{H}^1(S; \mathcal{G}) = \operatorname{colim}_{\mathcal{U}} \check{H}^1(\mathcal{U}; \mathcal{G})$ over refinement. For abelian $\mathcal{G}$ it is a group and $\check{H}^2$ is defined as usual.

3 Layer I: the information residual (recalled and sharpened)

Let $c : X \to Y$, $A_i : X \to Z_i$, $a = (A_i)_i : X \to Z = \prod_i Z_i$.

Theorem 3.1 (Singleton criterion)**.** [N] *There is $d : c(X) \to Z$ with $a = d \circ c$ iff $\mathcal{R}_y = \{a(x) : c(x) = y\}$ is a singleton for every $y \in c(X)$; then d is unique.*

Proof. If $a = d \circ c$ then $\mathcal{R}_y = \{d(y)\}$. Conversely set $d(y)$ to the unique element of $\mathcal{R}_y$. Uniqueness: c is surjective onto $c(X)$. □

Definition 3.2. A *sufficient encoding* is a surjection $e : X \to W$ with $c = u \circ e$, $a = v \circ e$ for some u, v. A morphism $e \to e'$ is $h : W \to W'$ with $e' = h \circ e$. Let $\mathbf{Suff}(\mathcal{K})$ be this category.

Theorem 3.3 (Terminal sufficient refinement)**.** [N] $r : X \to E = \{(c(x), a(x))\}$, $r(x) = (c(x), a(x))$, *is terminal in* $\mathbf{Suff}(\mathcal{K})$. *Hence it is unique up to unique isomorphism, and its kernel partition is the unique coarsest sufficient partition. It is natural: for $f : X' \to X$ with $c' = c \circ f$, $a' = a \circ f$, the inclusion $E' \hookrightarrow E$ satisfies $r \circ f = \iota \circ r'$.*

Proof. For a sufficient e, define $h(e(x)) = (u(e(x)), v(e(x))) = (c(x), a(x))$; it is well defined because u, v are functions on W, and unique because e is surjective. Terminal objects are unique up to unique isomorphism [17]. Kernel partitions of sufficient encodings are exactly the partitions refining $\ker c \cap \ker a$; the coarsest is $\ker r$. Naturality: $r(f(x')) = (c'(x'), a'(x')) = r'(x') \in E' \subseteq E$. □

Remark 3.4 (Relation to minimal sufficiency). Theorem 3.3 is the deterministic analogue of the Lehmann–Scheffé minimal sufficient statistic [14]: the coarsest partition through which every declared task factors. It is not the Blackwell order (Ladder R14), which compares garblings.

Proposition 3.5 (Joint witnesses are required)**.** [N] *Retaining the marginals $\{\mathcal{R}_y^{(i)}\}_i$ instead of $\mathcal{R}_y$ can strictly enlarge the possibility set: $\mathcal{R}_y \subseteq \prod_i \mathcal{R}_y^{(i)}$, with strict inclusion for $X = \{0, 1\}$, c constant, $A_1 = A_2 = \mathrm{id}$.*

Theorem 3.6 (Operational closure as a final behaviour)**.** [N] *Let M be a monoid of maps $X \to X$ and $O : X \to Z$. Define $\beta : X \to Z^M$, $\beta(x)(m) = O(m(x))$. Then $\ker \beta$ is the largest equivalence relation contained in $\ker O$ and stable under M; $X/\ker \beta \cong \beta(X)$, every $m \in M$ descends, and β is unique up to unique isomorphism among M-stable observation-preserving quotients through which all others factor.*

Proof. If $\beta(x) = \beta(y)$ then for $n \in M$, $\beta(nx)(m) = O(mnx) = O(mny) = \beta(ny)(m)$: stability. $\mathrm{id} \in M$ gives $\ker \beta \subseteq \ker O$. If $\approx$ is stable and inside $\ker O$, then $x \approx y$ implies $mx \approx my$, hence $O(mx) = O(my)$ for all m, so $\approx \subseteq \ker \beta$. This is the Myhill–Nerode congruence [21] and the greatest bisimulation [18]. □

Proposition 3.7 (Smooth descent)**.** [C: submersion] *If $c : X \to Y$ is a surjective submersion of manifolds and $A : X \to Z$ is smooth and constant on fibres, then the descended d is smooth. Without the submersion hypothesis this fails: $c(x) = x^3$, $A(x) = x$, $d = \sqrt[3]{\cdot}$ is not differentiable at 0 (A13).*

Proof. Local sections s of a submersion exist; $d = A \circ s$ locally. □

4 Layer II: RG data at a point and its automorphisms

Definition 4.1 (RG datum)**.** An RG datum of dimensions (p, q) is $(R, N, \Sigma_R, C, \Sigma_N)$ with $R \cong \mathbb{R}^p$, $N \cong \mathbb{R}^q$, $\Sigma = \left(\begin{smallmatrix} \Sigma_R & C \\ C^\top & \Sigma_N \end{smallmatrix}\right)$ positive definite on $(R \oplus N)^*$. In frames, $(A, B) \in \mathrm{GL}(p) \times \mathrm{GL}(q)$ acts by $\Sigma_R \mapsto A\Sigma_R A^\top$, $C \mapsto ACB^\top$, $\Sigma_N \mapsto B\Sigma_N B^\top$. The *whitened coupling* is $K = \Sigma_R^{-1/2} C \Sigma_N^{-1/2}$; its singular values are the *canonical correlations*.

Proposition 4.2 (The coupling is complement-relative)**.** [X: complement] *Let* $J = \left(\begin{smallmatrix} I_p \\ 0 \end{smallmatrix}\right)$ *and* $T = \left(\begin{smallmatrix} I & K' \\ 0 & I \end{smallmatrix}\right)$, $K' \in \mathbb{R}^{p \times q}$. *Then* $TJ = J$, *the Fisher pullback* $G = J^\top \Sigma^{-1} J$ *and the block* $D = \Sigma_N$ *are invariant under* $\Sigma \mapsto T\Sigma T^\top$, *while* $C \mapsto C + K'D$ *and* $L = CD^{-1} \mapsto L + K'$. *In particular* $C' = 0$ *for* $K' = -CD^{-1}$.

Proof. $T\Sigma T^\top$ has blocks $A' = A + K'C^\top + CK'^\top + K'DK'^\top$, $C' = C + K'D$, $D' = D$, and $A' - C'D^{-1}C'^\top = A - CD^{-1}C^\top$. Since $T^{-1}J = J$, $J^\top (T\Sigma T^\top)^{-1} J = (T^{-1}J)^\top \Sigma^{-1} (T^{-1}J) = G$. □

Remark 4.3. Proposition 4.2 (A12) shows that "$C_{RN} = 0$" (Ladder R9) and the coordinate L are meaningful only after the complement N is *declared*; L ranges over an affine space with no invariant origin. If N is declared as the Σ-orthogonal complement then $C = 0$ identically and the canonical correlations vanish; this is the convention of the Groupoid paper (GR04). The final print *Resolution Geometry* declares $N = \ker J^*$ for the observation metric, which satisfies H-split. Both conventions are admissible; they define different contracts and different remainders. We work with a declared N (H-split) and treat $C = 0$ as a special type.

A second contract choice concerns the acting group. If the observation metric is also part of the contract, morphisms are restricted to $\mathrm{O}(p) \times \mathrm{O}(q)$ in orthonormal frames. The pointwise moduli are then the orbit space of dimension $p + q + pq$ (final print, Proposition "Second-order quotient"), and canonical correlations no longer coordinate it. The global part is still classified by Theorem 5.3 with coefficients in the joint stabilizer. The explicit block-bundle form of Theorem 6.4 is stated, and claimed, only for the covariance-only contract $\mathrm{GL}(p) \times \mathrm{GL}(q)$.

Theorem 4.4 (Orbit and stabilizer of an RG datum)**.** [N]

(i) *Two RG data of dimensions* (p, q) *are isomorphic iff they have the same canonical correlations with multiplicity [9].*

(ii) *Let the distinct nonzero canonical correlations be* $\sigma_1 > \cdots > \sigma_k$ *with multiplicities* m_j, $r = \sum m_j$. *Then*

$$\mathrm{Aut}(R, N, \Sigma_R, C, \Sigma_N) \;\cong\; \prod_{j=1}^{k} \mathrm{O}(m_j) \times \mathrm{O}(p - r) \times \mathrm{O}(q - r).$$

(iii) *Consequently*

$$\dim \mathrm{Aut} = \sum_{j=1}^{k} \binom{m_j}{2} + \binom{p-r}{2} + \binom{q-r}{2}, \qquad \pi_0(\mathrm{Aut}) \cong (\mathbb{Z}/2)^\kappa,$$

where κ *is the number of factors of positive rank among* $m_1, \ldots, m_k, p - r, q - r$.

Proof. (i) Whitening by $(\Sigma_R^{-1/2}, \Sigma_N^{-1/2})$ reduces to $\Sigma_R = I$, $\Sigma_N = I$, $C = K$; the residual group is $\mathrm{O}(p) \times \mathrm{O}(q)$ acting by $K \mapsto AKB^\top$, whose orbits are classified by singular values (SVD). (ii) Conjugating by the whitening, the stabilizer is $\{(A, B) \in \mathrm{O}(p) \times \mathrm{O}(q) : AD = DB\}$ with $D = \left(\begin{smallmatrix} \Lambda & 0 \\ 0 & 0 \end{smallmatrix}\right)$, $\Lambda = \mathrm{diag}(\sigma_1 I_{m_1}, \ldots, \sigma_k I_{m_k})$. From $AD = DB$ and orthogonality, $D^\top A = BD^\top$, hence $ADD^\top = DBD^\top = DD^\top A$ and $BD^\top D = D^\top DB$. Thus A commutes with

$DD^\top = \mathrm{diag}(\sigma_1^2 I_{m_1}, \dots, \sigma_k^2 I_{m_k}, 0_{p-r})$ and is block diagonal $A = \mathrm{diag}(A_1, \dots, A_k, A_0)$; likewise $B = \mathrm{diag}(B_1, \dots, B_k, B_0)$. Then $AD = DB$ reads $\sigma_j A_j = \sigma_j B_j$, so $A_j = B_j$. Conversely every such pair stabilizes D. (iii) is immediate from (ii). □

Corollary 4.5 (Refutation of the rigidity proposition)**.** [N] *The Lock Conclusion states that* $\dim \mathrm{Aut}$ *is decreasing in* $\mathrm{rank}\, C$ *and that full-rank coupling rigidifies to the trivial automorphism. Both are false:*

(a) $p = q = 3$, $C = \frac{1}{2} I_3$ *(rank 3):* $\mathrm{Aut} \cong \mathrm{O}(3)$, $\dim = 3 > 2 = \dim \mathrm{Aut}$ *at rank 1;*

(b) $p = q = 3$, *distinct nonzero canonical correlations (rank 3):* $\mathrm{Aut} \cong (\mathbb{Z}/2)^3$, *which has dimension* 0 *but is not trivial.*

The values $6, 2, 0, 0$ *reported for ranks* $0, 1, 2, 3$ *are correct for* distinct *canonical correlations only.*

Remark 4.6 (Why the sweep missed it). The set of couplings with a repeated nonzero singular value has Lebesgue measure zero. A random sweep of 5000 Gaussian couplings (corpus, seed 3; reproduced here with seed 20266072) finds no counterexample, while the exact stratum $C = \frac{1}{2} I$ gives $\dim \mathrm{Aut} = 3$ (A05). Universal statements over a stratified space must be tested stratum by stratum; random sampling tests only the open stratum.

5 Layer III: families, stacks and twisted forms

Definition 5.1 (RG family)**.** An RG family of dimensions (p, q) over S is $y = (R, N, \Sigma_R, C, \Sigma_N)$: vector bundles R, N of ranks p, q and smooth sections $\Sigma_R \in \Gamma(\mathrm{Sym}^2 R)$, $\Sigma_N \in \Gamma(\mathrm{Sym}^2 N)$, $C \in \Gamma(R \otimes N)$ such that the fibre data are RG data. Its *multiplicity type* at s is $\mu(s) = (p, q; m_1, \dots, m_k)$ from Theorem 4.4. A morphism $y \to y'$ is a pair of bundle isomorphisms (α, β) over id_S with $\alpha \Sigma_R \alpha^\top = \Sigma'_R$, $\alpha C \beta^\top = C'$, $\beta \Sigma_N \beta^\top = \Sigma'_N$. $\mathrm{RG}_\mu(U)$ denotes families of constant type μ over U.

Proposition 5.2 (Local repairs form a stack)**.** [N] $U \mapsto \mathrm{RG}_\mu(U)$ *is a stack over* S.

Proof. (S1) Morphisms are pairs of bundle maps satisfying pointwise equations; bundle maps glue and the equations are local. (S2) Vector bundles glue along cocycles of bundle isomorphisms; tensor fields preserved by the transition maps glue to global sections; the type is pointwise. □

Theorem 5.3 (Twisted forms)**.** [N] *For a stack* $\mathcal{F}$ *and* $x \in \mathcal{F}(S)$ *there is a bijection of pointed sets, natural in* S, $\mathrm{Forms}(x) \cong \check{H}^1(S; \mathrm{Aut}_x)$.

Proof. Given a twisted form y, choose a cover and isomorphisms $\varphi_a : x|_{U_a} \to y|_{U_a}$; put $g_{ab} = \varphi_a^{-1} \varphi_b \in \mathrm{Aut}_x(U_{ab})$. Then $g_{ab} g_{bc} = g_{ac}$. Replacing φ_a by $\varphi_a h_a$ replaces g_{ab} by $h_a^{-1} g_{ab} h_b$; an isomorphism $y \cong y'$ composed with the φ_a gives the same cocycle; refinement restricts cocycles. So $[y] \mapsto [g]$ is well defined. Injectivity: if $g'_{ab} = h_a^{-1} g_{ab} h_b$, the local isomorphisms $\psi_a = \varphi'_a h_a^{-1} \varphi_a^{-1} : y|_{U_a} \to y'|_{U_a}$ agree on overlaps and glue by (S1). Surjectivity: a cocycle g is a descent datum $(x|_{U_a}, g_{ab})$; by (S2) it is effective, and the glued object is a twisted form with cocycle g. Naturality: pullback commutes with every step. See [6, 24, 27]. □

Proposition 5.4 (Reference dependence)**.** [X: reference model] $\mathrm{Forms}(x)$ *is a set; its base point is* $[x]$. *For a twisted form* y *there is a natural bijection* $\check{H}^1(S; \mathrm{Aut}_x) \cong \check{H}^1(S; \mathrm{Aut}_y)$ *sending* $[y]$ *to the base point. For non-abelian* Aut_x *there is in general no group structure, and the global automorphism groups of* x *and* y *may differ: over* S^1 *with* $\mathrm{Aut} = S_3$ *(discrete), the trivial form has* $\mathrm{Aut} = S_3$ *and the form with holonomy* (123) *has* $\mathrm{Aut} \cong \mathbb{Z}/3$ *(A11).*

Proof. The bijection sends a form z to the class of $\mathrm{Isom}(y, z)$ viewed as an Aut_y-torsor [24]. Global automorphisms of a flat form with holonomy h are the centralizer of h. □

Remark 5.5. "A nontrivial class" is therefore a relation between two objects, not a property of one, unless Aut is abelian, where $\check{H}^1$ is a group and differences of classes are intrinsic.

Proposition 5.6 (Discrete structure groups and holonomy)**.** [C: S connected, G discrete] *If* Aut_x *is the constant sheaf of a discrete group* G*, then* $\mathrm{Forms}(x) \cong \mathrm{Hom}(\pi_1(S, s_0), G)/G$ *(conjugation). In particular over a tree-like (contractible) base there is one class, over* S^1 *the classes are conjugacy classes of* G*, and over a wedge of two circles there are* $|G|^{-1} \sum_{g \in G} |C_G(g)|^2$ *classes (Burnside).*

The battery (A02) recomputes these numbers for $G = S_3$ by brute-force gauge orbits on nerves, independently of the proof: 1 (tree and filled simplex), 3 (circle), 11 (figure eight).

6 The Global Structure Theorem

For an RG family y define, in any local frames, the endomorphism fields

$$T_R = C\Sigma_N^{-1}C^\top\Sigma_R^{-1} \in \Gamma(\mathrm{End}\, R), \qquad T_N = C^\top\Sigma_R^{-1}C\Sigma_N^{-1} \in \Gamma(\mathrm{End}\, N),$$

$$M = C\Sigma_N^{-1} \in \Gamma(\mathrm{Hom}(N, R)),$$

and the metrics $g_R = \Sigma_R^{-1}$, $g_N = \Sigma_N^{-1}$.

Lemma 6.1 (Frame independence and spectral structure)**.** [N] T_R, T_N, M, g_R, g_N *are tensorial (independent of frames).* T_R *is* g_R*-self-adjoint with spectrum* $\{\sigma_j^2\} \cup \{0\}$*,* T_N *is* g_N*-self-adjoint with the same nonzero spectrum,* $T_R M = M T_N$*, and for* $v \in \ker(T_N - \sigma_j^2)$*,* $g_R(Mv, Mv) = \sigma_j^2 g_N(v, v)$*.*

Proof. Under (A, B): $C\Sigma_N^{-1}C^\top \mapsto A(C\Sigma_N^{-1}C^\top)A^\top$ and $\Sigma_R^{-1} \mapsto A^{-\top}\Sigma_R^{-1}A^{-1}$, so $T_R \mapsto AT_RA^{-1}$; similarly for T_N, and $M \mapsto AMB^{-1}$. In whitened frames $T_R = KK^\top$, $T_N = K^\top K$, $M = K$, whence the spectral statements and $T_R M = KK^\top K = MT_N$. Finally $g_R(Mv, Mv) = v^\top K^\top Kv = \sigma_j^2|v|^2$. □

Definition 6.2 (Block bundles)**.** For y of constant type μ let $E_j^R = \ker(T_R - \sigma_j^2)$, $K_R = \ker T_R$, $E_j^N = \ker(T_N - \sigma_j^2)$, $K_N = \ker T_N$, with the metrics induced by g_R, g_N, and $\Theta_j = \sigma_j^{-1} M|_{E_j^N} : E_j^N \to E_j^R$.

Lemma 6.3 (Smoothness)**.** [C: constant multiplicity] *If* μ *is locally constant, the functions* σ_j *are smooth and* E_j^R, K_R, E_j^N, K_N *are smooth subbundles;* Θ_j *is a smooth isometric bundle isomorphism.*

Proof. Near s_0 choose disjoint circles Γ_j around $\sigma_j(s_0)^2$ and Γ_0 around 0, each enclosing exactly one eigenvalue cluster for nearby s (continuity of the spectrum). The Riesz projections $P_j = \frac{1}{2\pi i}\oint_{\Gamma_j}(z - T_R)^{-1}dz$ are smooth, idempotent and of locally constant rank [11, Ch. II]; the image of a smooth constant-rank idempotent is a smooth subbundle. By constant multiplicity, $T_R P_j = \sigma_j^2 P_j$ and $\sigma_j^2 = \mathrm{tr}(T_R P_j)/m_j$ is smooth. Θ_j maps E_j^N into E_j^R by $T_R M = MT_N$ and is isometric by Lemma 6.1; equal ranks make it an isomorphism. □

Theorem 6.4 (Global Structure Theorem)**.** [N] *on* RG_μ *The functor*

$$\Phi_S(y) = \big(\sigma(y);\, E_1^R, \ldots, E_k^R,\, K_R,\, K_N\big), \qquad \Phi_S(\alpha, \beta) = \big(\alpha|_{E_1^R}, \ldots, \alpha|_{E_k^R},\, \alpha|_{K_R},\, \beta|_{K_N}\big)$$

is an equivalence of groupoids $\mathrm{RG}_\mu(S) \simeq C^\infty(S, \Delta_k) \times \prod_j \mathrm{Euc}_{m_j}(S) \times \mathrm{Euc}_{p-r}(S) \times \mathrm{Euc}_{q-r}(S)$*, compatible with pullback along smooth maps* $S' \to S$*.*

Proof. Well defined. A morphism (α, β) intertwines T_R with T'_R (by the transformation law of Lemma 6.1), preserves g_R, hence maps E_j^R isometrically onto $E_j'^R$ and K_R onto K'_R; similarly β on K_N. Since σ is an isomorphism invariant, morphisms exist only between families with equal σ.

Faithful. $R = \bigoplus_j E_j^R \oplus K_R$ (g_R-orthogonal eigen-decomposition), so α is determined by its restrictions. On $N = \bigoplus_j E_j^N \oplus K_N$, $\alpha M = M'\beta$ (equivalent to $\alpha C \beta^\top = C'$) gives $\beta|_{E_j^N} = \Theta_j'^{-1} \alpha|_{E_j^R} \Theta_j$, so β is determined by α and $\beta|_{K_N}$.

Full. Given isometries $\phi_j : E_j^R \to E_j'^R$, $\phi_R : K_R \to K'_R$, $\phi_N : K_N \to K'_N$, set $\alpha = \bigoplus \phi_j \oplus \phi_R$ and $\beta = \bigoplus \Theta_j'^{-1} \phi_j \Theta_j \oplus \phi_N$. These are g-isometries, hence preserve $\Sigma_R = g_R^{-1}$ and $\Sigma_N = g_N^{-1}$, and $\alpha M = M'\beta$ holds blockwise (both sides vanish on K_N).

Essentially surjective. Given $(\sigma; E_1, \dots, E_k, K'_R, K'_N)$ put $R = \bigoplus_j E_j \oplus K'_R$, $N = \bigoplus_j E_j \oplus K'_N$, g_R, g_N the orthogonal sums, $\Sigma_R = g_R^{-1}$, $\Sigma_N = g_N^{-1}$, and $M = \bigoplus_j \sigma_j \operatorname{id}_{E_j} \oplus 0$, $C = M\Sigma_N$. In orthonormal frames $\Sigma = \left(\begin{smallmatrix} I & M \\ M^\top & I \end{smallmatrix}\right)$, which is positive definite because $\|M\| = \sigma_1 < 1$; the canonical correlations are σ with multiplicities m_j. Its image under Φ_S is canonically isomorphic to the given tuple.

Base change. T_R, T_N, M and the Riesz projections commute with pullback. □

Corollary 6.5 (Classification)**.** [C: S paracompact; homotopy part on CW] *Isomorphism classes of RG families of type μ over S are in natural bijection with*

$$C^\infty(S, \Delta_k) \times \prod_j \operatorname{Vect}_{m_j}(S) \times \operatorname{Vect}_{p-r}(S) \times \operatorname{Vect}_{q-r}(S),$$

where $\operatorname{Vect}_m(S) \cong [S, B\,\mathrm{O}(m)] = [S, \operatorname{Gr}_m(\mathbb{R}^\infty)]$. *For constant σ the global part equals*

$$\check{H}^1\big(S; C^\infty(-, \operatorname{Aut}_\mu)\big) \cong [S, B \operatorname{Aut}_\mu].$$

Proof. Every vector bundle over a paracompact base admits a metric, and two metrics on the same bundle are isometric via the positive square root of their comparison endomorphism, so $\operatorname{Euc}_m(S)$ and $\operatorname{Vect}_m(S)$ have the same isomorphism classes. The identification with $[S, B\,\mathrm{O}(m)]$ is classical [25, 19, 10]; smooth and continuous classifications agree for finite-dimensional Lie structure groups [10, 20]. For constant σ, all families of type μ are twisted forms of one model and Theorem 5.3 applies with Aut_μ from Theorem 4.4. □

Corollary 6.6 (Consequences)**.** [N]

(a) ***Contractible base.*** *If S is contractible, every RG family of type μ is isomorphic to the constant family with the same σ. A nonzero global class is a property of a family over a topologically nontrivial base, not of RG.*

(b) ***Circle.*** *For constant σ, $|\operatorname{Forms}_\mu(S^1)| = |\pi_0(\operatorname{Aut}_\mu)| = 2^{\kappa(\mu)}$, each class determined by the first Stiefel–Whitney classes of the positive-rank blocks. For $p = q = 3$: $C = 0$: 4; rank 1:* 8*; rank 2 distinct:* 16*; rank 3 distinct:* 8*; rank 2 equal:* 8*; rank 3 with one equal pair:* 4*; $C = \frac{1}{2}I$:* 2 *(A17). The count is not monotone in rank.*

(c) ***Sphere.*** *Over S^2 a block of rank* 1 *contributes one class, of rank* 2 *classes indexed by $|e| \in \mathbb{N}$, of rank ≥ 3 two classes (w_2), since $[S^2, B\,\mathrm{O}(m)] = \pi_1(\mathrm{O}(m))/\pi_0(\mathrm{O}(m))$.*

(d) ***Dimension zero is not global rigidity.*** *Distinct full-rank canonical correlations give* $\dim \operatorname{Aut} = 0$ *yet* $2^{\min(p,q)}$ *classes over S^1 when $p = q$ (A06).*

(e) ***Local invariants are not complete.*** *The family over S^1 with $R_t = \operatorname{span}(\cos\frac{t}{2}, \sin\frac{t}{2})$, $N_t = R_t^\perp$ and constant canonical correlation $\rho \in (0, 1)$ has constant spectrum, constant canonical correlation and constant* $\operatorname{Aut} \cong \mathbb{Z}/2$*, but is not isomorphic to the constant family: R is the Möbius bundle (A03).*

Proof. (a) Bundles over contractible paracompact spaces are trivial and $C^\infty(S, \Delta_k)$ is unchanged. (b) $[S^1, BG]$ is the set of conjugacy classes of $\pi_0(G)$, here $(\mathbb{Z}/2)^\kappa$ by Theorem 4.4(iii). (c) Standard: $\pi_1(\mathrm{O}(1)) = 0$, $\pi_1(\mathrm{O}(2)) = \mathbb{Z}$ with π_0 acting by -1, $\pi_1(\mathrm{O}(m)) = \mathbb{Z}/2$ for $m \geq 3$. (d) is (b) with $\kappa = k = p$. (e) A nowhere-vanishing section $f(t)u(t)$ of R would satisfy $f(2\pi) = -f(0)$, contradicting the intermediate value theorem. □

Corollary 6.7 (Global Fisher blindness)**.** [N] *The Fisher tensor* $G = (\Sigma_R - C\Sigma_N^{-1}C^\top)^{-1}$ *satisfies* $G^{-1}\Sigma_R^{-1} = I - T_R$*. Hence* (R, Σ_R, G) *determines* σ*,* E_j^R *and* K_R*, and the isomorphism classes of families with prescribed* (R, Σ_R, G) *and prescribed* q *are in bijection with* $\mathrm{Vect}_{q-r}(S)$*, the class of the null-kernel bundle* K_N*. Over* S^1 *with* $p = 1$*,* $q = 2$*,* $r = 1$ *there are exactly two such families, with* K_N *trivial resp. Möbius (A18).*

Proof. $G^{-1} = \Sigma_R - C\Sigma_N^{-1}C^\top = (I - T_R)\Sigma_R$. The rest follows from Theorem 6.4: $(\sigma, E_\bullet, K_R)$ are fixed and K_N is free. □

Remark 6.8 (Ambient bundle as part of the contract). [X: contract] If the observation bundle $O = R \oplus N$ is fixed in advance (e.g. trivial), an RG family with that ambient is a reduction of the frame bundle of O to the stabilizer of a model datum, i.e. a section of the associated bundle with fibre $\mathrm{GL}(p+q)/\mathrm{Stab}$ [25, 10]. The remainder then changes (for instance $R \oplus N$ must be isomorphic to O). Corollary 6.6(e) lives in this variant. Residual structure is contract-dependent; the classification must be stated for a declared contract.

7 Universality and uniqueness of the remainder

Definition 7.1 (Complete natural invariant)**.** A *complete natural invariant* of RG families of type μ is a functor $\mathcal{I} : \mathbf{Man}^{\mathrm{op}} \to \mathbf{Set}$ together with elements $\iota_S(y) \in \mathcal{I}(S)$ such that (a) $y \cong y' \Rightarrow \iota_S(y) = \iota_S(y')$, (b) $\iota_{S'}(f^*y) = \mathcal{I}(f)(\iota_S(y))$, (c) $\iota_S(y) = \iota_S(y') \Rightarrow y \cong y'$.

Theorem 7.2 (Universality)**.** [N] *Let* $\mathcal{I}_0(S) = \pi_0\mathrm{RG}_\mu(S)$ *with the tautological invariant. For every complete natural invariant* $(\mathcal{I}, \iota)$ *there is a unique natural injection* $\lambda : \mathcal{I}_0 \to \mathcal{I}$ *with* $\iota = \lambda \circ [\,\cdot\,]$*. By Theorem 6.4,* $\mathcal{I}_0 \cong C^\infty(-, \Delta_k) \times \prod_j \mathrm{Vect}_{m_j} \times \mathrm{Vect}_{p-r} \times \mathrm{Vect}_{q-r}$*; this initial invariant is unique up to unique natural isomorphism. For fixed* σ *the functor* $\pi_0\mathrm{RG}_\mu(-)$ *on CW complexes is represented by* $B\,\mathrm{Aut}_\mu \simeq \prod_j B\,\mathrm{O}(m_j) \times B\,\mathrm{O}(p-r) \times B\,\mathrm{O}(q-r)$*, and a representing object is unique up to homotopy equivalence.*

Proof. Define $\lambda([y]) = \iota(y)$: well defined by (a), injective by (c), natural by (b); unique because $[\,\cdot\,]$ is surjective. Initial objects are unique up to unique isomorphism. Representability is Corollary 6.5 with $B(G_1 \times G_2) \simeq BG_1 \times BG_2$; uniqueness of a representing object follows from the Yoneda lemma in the homotopy category. □

Remark 7.3 (What uniqueness does and does not say). Uniqueness is of the *remainder object*: any complete natural invariant contains exactly the information $(\sigma, [E_\bullet], [K_R], [K_N])$, presented perhaps differently. It does not say that $\mathrm{Vect}_m(S)$ has a simple complete set of computable invariants; Stiefel–Whitney and Pontryagin classes are not complete in general. That is a classical, not an RG, question.

8 The second level: gluing defects, holonomy, covers

Proposition 8.1 (Defects with an abelian band)**.** [C: abelian band] *Let* $x_a \in \mathcal{F}(U_a)$ *be pairwise locally isomorphic objects of a stack whose automorphism sheaves are abelian and canonically identified with a sheaf* $\mathcal{A}$*. Choose isomorphisms* $\varphi_{ab} : x_b|_{U_{ab}} \to x_a|_{U_{ab}}$ *and put* $c_{abc} = \varphi_{ab}\varphi_{bc}\varphi_{ca} \in \mathcal{A}(U_{abc})$*. Then* c *is a Čech 2-cocycle; changing* φ_{ab} *to* $\varphi_{ab}t_{ab}$ *changes* c *by* δt*; the class* $[c] \in \check{H}^2(\mathcal{U}; \mathcal{A})$ *is independent of all choices; and the* φ_{ab} *can be re-chosen to satisfy the cocycle*

condition (so that the x_a glue) iff $[c] = 0$. If the transitions are rigid *($\mathcal{A}$ trivial), $c \neq 1$ is a plain local inconsistency.*

Proof. On U_{abcd}, commutativity of $\mathcal{A}$ gives $c_{bcd}c_{acd}^{-1}c_{abd}c_{abc}^{-1} = 1$ by expanding each c and cancelling $\varphi_{xy}\varphi_{yx}$. Re-choosing multiplies c by $t_{ab}t_{bc}t_{ca} = (\delta t)_{abc}$. If $[c] = 0$, $c = \delta t$ and $\varphi_{ab}t_{ab}^{-1}$ satisfies the cocycle condition; the stack glues. Conversely a cocycle choice gives $c = 1$. Independence from the x_a up to isomorphism follows by transporting choices. See [6, 4, 5]. □

Example 8.2 (Removable defect; A08a)**.** On a filled triangle with $\mathcal{A} = \mathbb{Z}/2$, the transitions $g_{01} = g_{12} = 1$, $g_{02} = -1$ have defect -1. Re-choosing $g_{02} \mapsto +1$ removes it; all 8 assignments are repairable since $\check{H}^2(\Delta^2; \mathbb{Z}/2) = 0$. The Groupoid paper's "admissible edges can still carry $\Omega = -I$" (GR06) is therefore not an obstruction when $-I$ belongs to the automorphism group of the local objects, which it does ($-I \in \mathrm{O}(r) \times \mathrm{O}(q)$).

Example 8.3 (Non-removable defect; A08c)**.** Let $S = \mathbb{R}P^2$ with its 6-vertex triangulation and the star cover (a Leray cover). Then $H^1 = H^2 = \mathbb{Z}/2$. Transitions given modulo the central subgroup $2\mathbb{Z}/4 \subset \mathbb{Z}/4$ by the nontrivial class $w \in H^1(\mathbb{R}P^2; \mathbb{Z}/2)$ have lifting defect $\beta(w) = w^2 \neq 0 \in H^2(\mathbb{R}P^2; \mathbb{Z}/2)$; the battery checks all 2^{15} lifts and all 2^{15} coboundaries. No re-choice glues.

Proposition 8.4 (Holonomy invariants)**.** [X: frame] *Holonomy of a flat structure (or a connection) around a based loop is defined up to conjugation by the structure group; its invariants are the conjugacy class, in particular the spectrum, and norms induced by a structure-invariant inner product. Matrix norms in arbitrary frames are not invariant: a rotation H has $\|H\|_F = \sqrt{2}$, while PHP^{-1} with $P = \left(\begin{smallmatrix} 1 & 3 \\ 0 & 1 \end{smallmatrix}\right)$ has $\|PHP^{-1}\|_F \approx 10.61$ (A09).*

Remark 8.5. Reported holonomy "norms $4.06, 4.29, 4.00$" (GR06) are evidence only if computed in a Σ-orthonormal frame or replaced by spectra.

Proposition 8.6 (Covers)**.** [C: Leray cover] *$\check{H}^\bullet(\mathcal{U}; \mathcal{A}) \to \check{H}^\bullet(S; \mathcal{A})$ is an isomorphism for locally constant $\mathcal{A}$ if every finite intersection of members of $\mathcal{U}$ is a disjoint union of contractible components and cochains are taken componentwise. A "nerve" that assigns one simplex per nonempty intersection is wrong otherwise: three 250° arcs on S^1 have disconnected intersections; componentwise Čech gives $H^1 = \mathbb{Z}/2$, the naive nerve (a filled triangle) gives 0 (A10).*

Proof. Leray's theorem for acyclic covers; the nerve theorem when all intersections are contractible [3, Ch. II], [7, §4.G]. □

9 The boundary: walls

Proposition 9.1 (Isotropy jump)**.** [N] *(negative) If y is a twisted form of a model x on U, then $\mathrm{Aut}(y_s) \cong \mathrm{Aut}(x_s)$ for all $s \in U$. Hence if the multiplicity type changes at s_0, no neighbourhood of s_0 carries a twisted-form description. Example: $p = q = 2$, $C(t) = \mathrm{diag}(t, 0)$: $\dim \mathrm{Aut} = 2$ at $t = 0$ and 0 for $t \neq 0$ (A07). Isotropy is upper semicontinuous: the wall group is larger.*

Theorem 9.2 (Forced wall monodromy)**.** [C: transversal simple collision] *Let y be an RG family over an open $U \subseteq S$ and $W \subset U$ a submanifold of codimension 2 such that on $U \setminus W$ two simple canonical correlations $\sigma_a > \sigma_b$ are separated from the rest of the spectrum and collide exactly on W, transversally (the traceless part of T_R restricted to the merged rank-2 block vanishes to first order along W with surjective normal derivative). Then on every small loop γ linking W once, $w_1(E_a^R|_\gamma) = w_1(E_b^R|_\gamma) \neq 0$, while the merged block $E_a^R \oplus E_b^R$ extends smoothly across W.*

Proof. The Riesz projection onto the cluster $\{\sigma_a^2, \sigma_b^2\}$ is smooth across W [11], giving a rank-2 subbundle F over a neighbourhood V of a point of W; take V contractible and an orthonormal

frame of F. In that frame $T_R|_F = \frac{1}{2}\operatorname{tr}(T_R|_F)I + \left(\begin{smallmatrix} a & b \\ b & -a \end{smallmatrix}\right)$ with $(a,b) : V \to \mathbb{R}^2$ vanishing exactly on W, and transversality makes $(a,b)|_\gamma$ a degree ± 1 map to $\mathbb{R}^2 \setminus 0$. Writing $(a,b) = \rho(\cos\phi, \sin\phi)$, the eigenline of the larger eigenvalue is spanned by $(\cos\frac{\phi}{2}, \sin\frac{\phi}{2})$, which returns as its negative after one turn of ϕ. Hence $E^R_a|_\gamma$ is the Möbius bundle; $E^R_b|_\gamma$ is its orthogonal complement in the trivial $F|_\gamma$ and has the same w_1. This is the real form of the von Neumann–Wigner codimension-2 crossing [28] and of the Herzberg–Longuet-Higgins sign change [8]. □

Example 9.3 (A15)**.** $p = q = 2$, $\Sigma_R = \Sigma_N = I$, $K(t,s) = \frac{1}{2}I + \frac{1}{5}\left(\begin{smallmatrix} t & s \\ s & -t \end{smallmatrix}\right)$ over $\mathbb{R}^2$. The canonical correlations $\frac{1}{2} \pm \frac{1}{5}\sqrt{t^2+s^2}$ collide only at the origin. The top singular line bundle returns with sign -1 around a loop enclosing the origin and $+1$ around a loop that does not.

Open problem 9.4 (Stratified classification). Classify RG families whose multiplicity type varies, in terms of per-stratum data (Theorem 6.4) together with specialization data along exit paths (merging and splitting of block bundles, rank change $\sigma \to 0$). The natural conjecture is an equivalence with a constructible stack, i.e. a functor from an exit-path category of the type stratification [26, 16]. Theorem 9.2 shows that any such classification must carry monodromy around codimension-2 collision loci. Nothing beyond this is proved here.

10 Exclusion of a primitive

Definition 10.1 ($\mathfrak{T}$-derived)**.** Let $\mathfrak{T}$ be the toolkit of established constructions: finite limits and quotients of sets; kernel pairs and coimages; greatest fixed points of monotone operators on partitions; smooth vector bundles, tensor fields, Riesz spectral projections; stacks, nonabelian $\check{H}^1$, abelian $\check{H}^2$; classifying spaces. A remainder attached to declared data is $\mathfrak{T}$-*derived* if it is the value of a construction in $\mathfrak{T}$ on those data and is characterized by a universal property. It is a *new primitive* if it is not $\mathfrak{T}$-derived.

Theorem 10.2 (No primitive)**.** [C: H-split, H-type, H-mor; abelian band for level two] *On the domain of the Main Theorem every component of the RG remainder is* $\mathfrak{T}$*-derived:*

Component	*Construction*	*Universal property*
information residual	$\mathcal{R}_y$, $E = \operatorname{im}(c,a)$	*terminal sufficient refinement (Thm. 3.3)*
operational residual	$\ker\beta$, $\beta : X \to Z^M$	*greatest stable congruence (Thm. 3.6)*
pointwise symmetry	Aut_μ	*stabilizer (Thm. 4.4)*
local moduli	$\sigma \in C^\infty(S, \Delta_k)$	*complete orbit invariant (Thm. 4.4(i))*
global class	$[E_\bullet, K_R, K_N]$	*initial complete natural invariant (Thm. 7.2)*
gluing obstruction	$[c] \in \check{H}^2(S; \mathcal{A})$	*gerbe class (Prop. 8.1)*

Proof. Each line is the cited result. □

Remark 10.3 (Honesty about the strength of this theorem). Within a set-theoretic contract, any remainder defined from the data is trivially a set-theoretic construction, so "no primitive" is formally cheap. The content of Theorem 10.2 is not definability but *identification*: each component is a named classical object with a universal property, computable invariants (Theorem 4.4(iii), Corollary 6.6), and no redundancy (Main Theorem (3)). Outside the domain, across walls, the remainder is open (Problem 9.4); *open* is not *primitive*, and the witnesses found there (Theorem 9.2) are again $\mathfrak{T}$-derived.

Proposition 10.4 (The no-faithful-functor verdict is false)**.** [N] *The forgetful functor* $\mathrm{RG}_\mu(S) \to \mathbf{VB}(S) \times \mathbf{VB}(S)$, $y \mapsto (R, N)$, *is faithful; composed with global sections it is faithful into* **Set**. *More strongly,* $\mathrm{RG}_\mu(S)$ *is equivalent to a product of classical groupoids (Theorem 6.4).*

Proof. Morphisms of RG families are pairs of bundle maps satisfying equations; distinct pairs have distinct images. The battery enumerates a finite $\mathbb{F}_3$ instance (A01). □

Remark 10.5. The meaningful content behind "no single foundation captures RG" is that its remainder is spread over several branches of §1.3. The Main Theorem shows these pieces assemble into one classical classification on constant-type strata. The residual Lock Conclusion verdict "nonzero descent class" is refuted by Corollary 6.6(a) and replaced by Corollary 6.5.

11 Proof of the Main Theorem

(1) is Theorem 6.4 with Corollary 6.5. (2) is Theorem 7.2. (3): Corollary 6.6(e) gives equal moduli with different global classes; over a point, $\Sigma_0 = I$ ($\sigma = \varnothing$) and $\Sigma_1 = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ ($\sigma = 1/\sqrt{2}$) have different moduli and no global data (A04). (4) is Theorem 10.2. (5) is Proposition 8.1. (6) is Proposition 9.1 and Theorem 9.2. □

12 Natural versus conditional ledger

Result	**Tag**	**Hypothesis or choice; what fails without it**
Singleton criterion, terminal refinement	N	set level only; smoothness needs Prop. 3.7
Operational closure	N	M a monoid of total maps; partial operations untreated
Smooth descent	C	submersion; x^3 counterexample
Coupling block C, coordinate L	X	declared complement; $L \mapsto L + K'$
Fisher pullback G, block D	N	invariant under complement change fixing J
Stabilizer theorem	N	none beyond Definition 4.1
Stack property	N	morphisms over id_S
Twisted forms $\leftrightarrow \check{H}^1$	N	as a set; base point is X (reference model)
Global Structure Theorem	N	on RG_μ, i.e. C (constant multiplicity)
Classification by $[S, B\,\mathrm{O}]$	C	paracompact base; CW for representability
Universality	N	complete natural invariants only
Gerbe class $\check{H}^2$	C	abelian band; non-abelian case needs Giraud bands
Holonomy	X	frame; promoted to conjugacy class
Čech computations	C	Leray cover
Wall monodromy	C	transversal simple collision
No primitive	C	H-split, H-type, H-mor

13 Corrections to the corpus

Table 3: Corpus claims attacked, with status and replacement.

ID	**Claim**	**Source**	**Status**	**Replacement**
A01	no faithful functor into any single foundation	Lock v:whole Concl.	refuted	Prop. 10.4
A02	residual is a nonzero descent class	Lock v:whole Concl.	refuted	Cor. 6.6(a)

ID	Claim	Source	Status	Replacement
A03	local similarity type complete	Global_RG §4	sharpened	Cor. 6.6(e)
A04	global data exhaust the remainder	implicit	refuted	Main Thm (3)
A05	dim Aut decreasing in rank; full rank $\Rightarrow$ trivial Aut	Lock Concl. prop:rigidity	refuted	Thm. 4.4
A06	dim Aut $= 0$ means rigid	Lock Concl. Interlock B	refuted	Cor. 6.6(d)
A07	principal Aut-bundle across types	Global_RG §2	boundary	Prop. 9.1
A08	Čech defect $\Omega = -I$ is a gluing failure	Groupoid GR06	sharpened	Prop. 8.1
A09	holonomy norms certify admissibility	Groupoid GR06	sharpened	Prop. 8.4
A10	naive nerve models of gluing	conversion bundle	boundary	Prop. 8.6
A11	nontrivial class intrinsic	Global_RG	boundary	Prop. 5.4
A12	$C_{RN} = 0$ intrinsic; (G, L, D) natural	Ladder R9	boundary	Prop. 4.2
A13	smooth descent of fibre-constant maps	Ladder R2 (smooth use)	boundary	Prop. 3.7
A14	terminal refinement unique and natural	residual lock	confirmed	Thm. 3.3
A15	strata glue without residual	Ladder R21	sharpened	Thm. 9.2
A16	global structure theorem (numerical)	this paper	confirmed	Thm. 6.4
A17	coupling rank controls global remainder	derived	sharpened	Cor. 6.6(b)
A18	Fisher blindness is only local	Ladder R9	sharpened	Cor. 6.7

14 The extended Theorem Ladder

The companion file `RG_-_Theorem_Ladder_v2.tex` adds Stratum 7 and two gate corrections. In the final print *Resolution Geometry* the same content appears as rungs L116–L123, with the open classification across walls as L126. The holding structure is:

Rung	Statement	Type	Depends on
R9′	Fisher blind to the null block at $C_{RN} = 0$ *relative to a declared complement*; globally the Fisher-relative remainder is $[K_N] \in \mathrm{Vect}_{q-r}(S)$	no-go (gated: H-split)	R8, R35
R21′	Stratified realization: across a codim-2 simple collision, $w_1 \neq 0$ is forced around the wall	gated / open globally	R17–R20, R41
R30	Task residual $\mathcal{R}_y$; descent iff singletons	forced (contract)	R0–R2
R31	Terminal sufficient refinement E; unique, natural	forced	R30
R32	Joint witnesses required (marginals lose correlation)	no-go	R30
R33	Operational closure = greatest stable congruence	imported	R31
R34	Smooth descent iff submersion (else counterexample)	gated (regularity)	R31
R35	Stabilizer $\prod \mathrm{O}(m_j) \times \mathrm{O}(p-r) \times \mathrm{O}(q-r)$; dimension and π_0 formulas	imported	R7
R36	RG families of constant type form a stack	forced (H-mor)	R35
R37	Twisted forms $\leftrightarrow \check{H}^1(S; \mathrm{Aut}_x)$, pointed by reference	imported	R36
R38	Global Structure Theorem: $\mathrm{RG}_\mu \simeq C^\infty(\Delta_k) \times \prod \mathrm{Euc}$	proved (gated: H-type)	R35–R37

Rung	Statement	Type	Depends on
R39	Universality: initial complete natural invariant; $B\,\mathrm{Aut}_\mu$ unique up to $\simeq$	proved	R38
R40	Second level: defect class in $\check{H}^2$ (abelian band)	imported (gated)	R36
R41	Isotropy jump $\Rightarrow$ no single-model description near a wall	no-go	R35
R42	Classification across walls (exit-path / constructible stack)	open	R38, R41
R43	No primitive: every remainder component is $\mathfrak{T}$-derived	proved (gated: H-split, H-type, H-mor)	R30–R40
R44	"No faithful functor" and "intrinsic nonzero descent class"	refuted	R38

15 The hostile falsification battery

`rg_final_remainder_attacks.py` (Python 3, NumPy, SymPy; master seed 20261022, per-attack seed $20261022 + 101k$) runs 18 attacks and 4 canaries. A manifest with the source hash is written before evaluation; results record the manifest hash, library versions and every witness. Exact arithmetic is used where a claim is algebraic (SymPy rationals, $\mathbb{F}_3$, $\mathbb{Z}/2$, $\mathbb{Z}/4$, Cayley-transform rational orthogonal matrices); floating point is used only for topological witnesses (frame transport on loops), where the decisive quantity is a sign (± 1 to 10^{-9}). Brute force replaces proofs where feasible: gauge orbits on nerves (46 656 cochains for the figure eight), all 2^{15} lifts on $\mathbb{R}P^2$, all partitions of a 4-point set with 82 944 naturality checks. Canaries (false statements: $H^1(S^1;\mathbb{Z}/2)$ trivial; $\dim \mathrm{Aut}(\frac{1}{2}I_3) = 0$; distinct full rank has trivial Aut; L invariant under complement change) are all rejected. Two fresh-process runs give identical witnesses.

Limitations. The battery is not blind: claims and fixtures share an author. Finite enumeration supports, but does not replace, the proofs. Numerical transport is used only for sign witnesses. No literature novelty search is claimed; references are classical and were checked from the author's knowledge, not against an online database.

16 Non-claims

1. No claim of novelty for any classical theorem used; the contribution is the assembly, the typing, and the corrections.
2. No classification across walls; Problem 9.4 is open.
3. No claim about contracts other than the declared one; changing the contract (ambient O fixed, N declared orthogonal, weaker tasks) changes the remainder.
4. No physical claim; the remainder is structural.
5. "No primitive" is relative to $\mathfrak{T}$ and to the hypotheses H-split, H-type, H-mor.

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